

MID-TERM EXAM (NOVEMBER, 7, 2015)

Problem I: Labor Supply and Government Spending
(12pts + 2 bonus points)

Bonus
14 pts + 0.5 + 0.5 = 15 pts

Consider a two-period production economy with three types of agents: 1) Households, 2) Firms and 3) Government. The economy is deterministic.

The inter-temporal utility of the representative household is given by:

$$\log(c_1) + \beta \log(c_2) - \beta n_2$$

where $\beta \in (0, 1)$ is the discount factor. Here, we assume that the representative household can only supply labor (n_2) in second period at a wage w . The household receives an endowment $\bar{y} > 0$ in first period, but no endowment in the second period. The representative household owns the firm and we define the intertemporal budget constraint as follows. Let Π be profits that the representative household receives from owning the firm. These profits are taken as given by the household. The household must also pay lump-sum taxes in first and second periods (denoted t_1, t_2). Then the household chooses consumptions c_1 and c_2 (at given prices p_1 and p_2) and labor supply n_2 in second period (at a wage denoted w) to maximize the utility subject to the intertemporal budget constraint

$$p_1 c_1 + p_2 c_2 = p_1 \bar{y} - p_1 t_1 + w n_2 - p_2 t_2 + \Pi$$

The firm uses the labor input n_2 and pays a wage w . The firm's production function is given by:

$$y_2 = n_2$$

The firm chooses optimally the level of labor (n_2) in period 2 such that she maximizes the profit:

$$\Pi = p_2 y_2 - w n_2$$

where p_2 is the price of output in period 2 and w the nominal wage.

Finally, the government consumes g_2 in second period (at a price p_2) and collects taxes t_1 and t_2 (notice that $g_1 = 0$). In addition, the government must satisfy its intertemporal budget constraint

$$p_2 g_2 = p_1 t_1 + p_2 t_2$$

It follows that the market clearing condition on goods market in period 1 is

$$c_1 = \bar{y},$$

and in period 2

$$c_2 + g_2 = y_2 (= n_2)$$

1. Define a competitive equilibrium of this economy (4pts). *+0.5 Bonus*
2. Solve for a competitive equilibrium. First solve the household problem (i.e. find the optimal consumptions c_1, c_2 and labor supply n_2 , for given prices). Next, determine the optimal labor demand, for given prices. What can you say about profits? After, use the equilibrium conditions on goods market (in period 1 and 2), on labor market and combine them with the optimality conditions in order to obtain the equilibrium allocations and prices. (4pts). *+0.5 Bonus*
3. Solve the central planner problem and compare the obtained solution to the one of the competitive equilibrium. Comment (2pts).
4. What is the effect of an increase in public spending g_2 on output, consumption and employment (in period 2) ? Comment. (2pts)
5. Show that the Ricardian equivalence holds in this economy (2 Bonus pts).

Exercise II: Tax Shocks and Stochastic Permanent Income (8 pts + 2 bonus points)

A given household seeks to maximize

$$E_t \sum_{i=0}^{\infty} \beta^i U(C_{t+i}) \quad \text{where the discount factor satisfies } \beta \in (0, 1)$$

under the budget constraint

$$A_{t+1} = (1 + r)A_t + Y_t^d - C_t$$

where C the real consumption, A the financial wealth and Y^d the (stochastic) real disposable labor income. The real interest rate r is strictly positive and constant. E_t is the conditional expectations operator. The instantaneous utility $U(C_t)$ is given by:

$$U(C_t) = \alpha_0 C_t - \frac{\alpha_1}{2} C_t^2$$

where α_0, α_1 are strictly positive and constant. In addition, we assume $\beta(1 + r) = 1$

1. Determine the optimal decision on consumption (or find the Euler equation on consumption) (3 pts).
2. Define the intertemporal budget constraint and thus determine the consumption decisions in terms of financial A_t and non-financial wealth (the expected discounted sum of the disposable labor income Y_t^d) (3 pts).
3. Now assume that the disposable labor is given by $Y_t^d = \bar{Y}(1 - \tau_t)$, where $\bar{Y} > 0$ and τ_t is a (stochastic) tax rate on the labor income. The process of τ_t is given by

$$\tau_t = \rho \tau_{t-1} + \varepsilon_t$$

where $|\rho| \leq 1$. The random term ε_t satisfies: i) the mean $E(\varepsilon_t) = 0$ and ii) the variance $V(\varepsilon_t) = \sigma_\varepsilon^2$. This shocks is also iid and unpredictable, i.e. $E_t \varepsilon_{t+i} = 0$ for all $i \geq 1$. Determine the consumption function, i.e. express the current consumption as a linear function of financial wealth A and stochastic tax rate τ (and a constant term) (2 pts).

4. Discuss the effects of the parameter ρ on the consumption function (2 Bonus pts).